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Java SE 27 & JDK 27

Class ObjectStreamField

java.lang.Object
    java.io.ObjectStreamField

All Implemented Interfaces:

Comparable<Object>

```
public class ObjectStreamField
extends Object
implements Comparable<Object>
```

A description of a Serializable field from a Serializable class. An array of ObjectStreamFields is used to declare the Serializable fields of a class.

Since:

1.2

See Also:

ObjectStreamClass

Constructor Summary

Constructors

Constructor	Description
ObjectStreamField(String name, Class<?> type)	Create a Serializable field with the specified type.
ObjectStreamField(String name, Class<?> type, boolean unshared)	Creates an ObjectStreamField representing a serializable field with the given name and type.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description
int	**compareTo**(Object obj)	Compare this field with another ObjectStreamField.
String	**getName**()	Get the name of this field.
int	**getOffset**()	

		Offset of field within instance data.
Class<?>	**getType**()	Get the type of the field.
char	**getTypeCode**()	Returns character encoding of field type.
String	**getTypeString**()	Return the JVM type signature.
boolean	**isPrimitive**()	Return true if this field has a primitive type.
boolean	**isUnshared**()	Returns boolean value indicating whether or not the serializable field represented by this ObjectStreamField instance is unshared.
protected void	**setOffset**(int offset)	Offset within instance data.
String	**toString**()	Return a string that describes this field.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, wait, wait, wait

Constructor Details

ObjectStreamField

```
public ObjectStreamField(String name,
                         Class<?> type)
```

Create a Serializable field with the specified type. This field should be documented with a `serialField` tag.

Parameters:

`name` - the name of the serializable field

`type` - the `Class` object of the serializable field

ObjectStreamField

```
public ObjectStreamField(String name,
                         Class<?> type,
                         boolean unshared)
```

Creates an ObjectStreamField representing a serializable field with the given name and type. If unshared is false, values of the represented field are serialized and deserialized in the default manner--if the field is non-primitive, object values are serialized and deserialized as if they had been written and read by calls to writeObject and readObject. If unshared is true, values of the represented field are serialized and deserialized as if they had been written and read by calls to writeUnshared and readUnshared.

Parameters:

`name` - field name

`type` - field type

`unshared` - if false, write/read field values in the same manner as writeObject/readObject; if true, write/read in the same manner as writeUnshared/readUnshared

Since:

1.4

Method Details

getName

```
public String getName()
```

Get the name of this field.

Returns:

a `String` representing the name of the serializable field

getType

```
public Class<?> getType()
```

Get the type of the field. If the type is non-primitive and this `ObjectStreamField` was obtained from a deserialized `ObjectStreamClass` instance, then `Object.class` is returned. Otherwise, the `Class` object for the type of the field is returned.

Returns:

a `Class` object representing the type of the serializable field

getTypeCode

```
public char getTypeCode()
```

Returns character encoding of field type. The encoding is as follows:

```
B            byte
C            char
D            double
F            float
I            int
J            long
L            class or interface
S            short
Z            boolean
[            array
```

Returns:

the typecode of the serializable field

getTypeString

```
public String getTypeString()
```

Return the JVM type signature.

Returns:

null if this field has a primitive type.

getOffset

```
public int getOffset()
```

Offset of field within instance data.

Returns:

the offset of this field

See Also:

`setOffset(int)`

setOffset

```
protected void setOffset(int offset)
```

Offset within instance data.

Parameters:

`offset` - the offset of the field

See Also:

`getOffset()`

isPrimitive

```
public boolean isPrimitive()
```

Return true if this field has a primitive type.

Returns:

true if and only if this field corresponds to a primitive type

isUnshared

```
public boolean isUnshared()
```

Returns boolean value indicating whether or not the serializable field represented by this ObjectStreamField instance is unshared.

Returns:

`true` if this field is unshared

Since:

1.4

compareTo

```
public int compareTo(Object obj)
```

Compare this field with another `ObjectStreamField`. Return -1 if this is smaller, 0 if equal, 1 if greater. Types that are primitives are "smaller" than object types. If equal, the field names are compared.

Specified by:

`compareTo` in interface `Comparable<Object>`

Parameters:

`obj` - the object to be compared.

Returns:

a negative integer, zero, or a positive integer as this object is less than, equal to, or greater than the specified object.

toString

```
public String toString()
```

Return a string that describes this field.

Overrides:

`toString` in class `Object`

Returns:

a string representation of the object

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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